

# GenQuantis Discovery Platform

Advanced Molecular Analysis & Formulation Report

Generated by: Raj | 2026-05-29 12:26

## LEAD OPTIMIZATION REPORT

**Target ID:** EGFR\_T790M  
**AI Strategy:** GeneticAlgorithm-RealTime  
**Status:** Completed

### Potency Optimization Results

|                     |                                 |
|---------------------|---------------------------------|
| Original pIC50      | 5.12                            |
| Optimized pIC50     | 5.12                            |
| Potency Improvement | 0.0%                            |
| Optimized SMILES    | CN1C=NC2=C1C(=O)N(C(=O)N2C)C... |

### Structure-Activity Relationship (SAR) Analysis

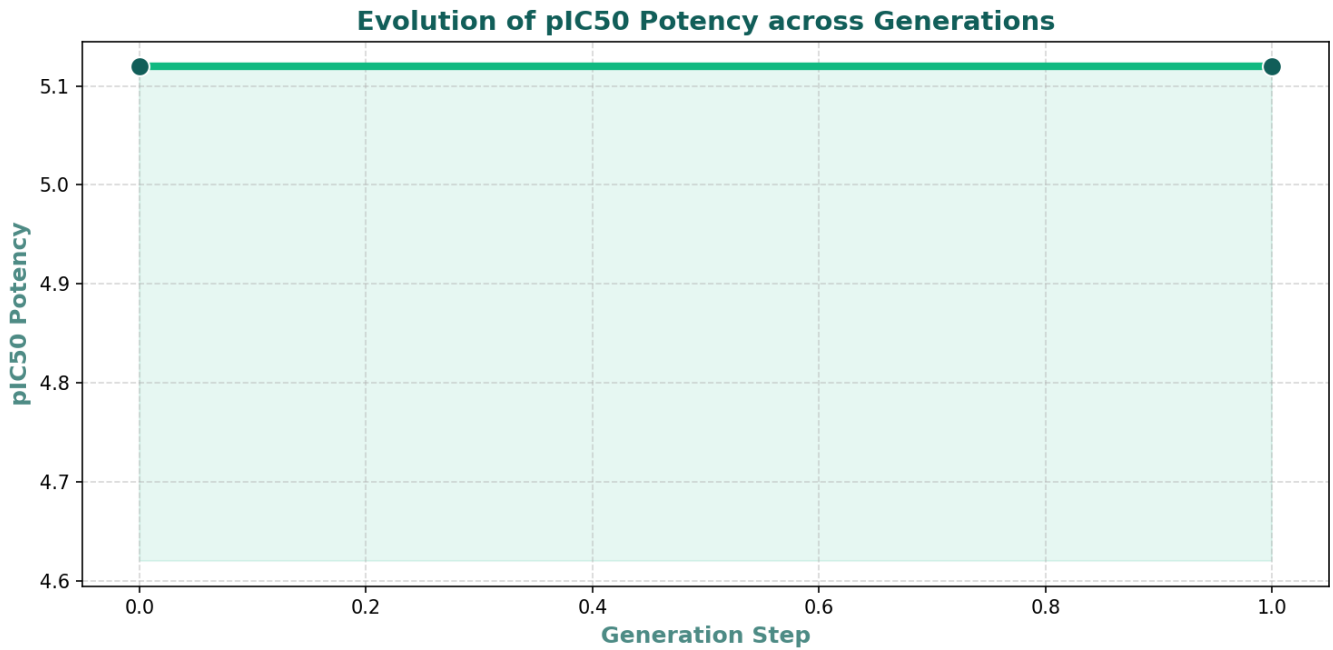
| Mutation / Modification       | Delta Affinity | Impact  |
|-------------------------------|----------------|---------|
| No successful mutations found | 0.0            | Neutral |

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## Generative Engine Real-Time Trajectory



*Report Note: Lead optimization scores are predicted using an XGBoost-based binding affinity model combined with evolutionary/generative AI strategies.*